

Variables

If you want to write useful code, you'll need to be able to store and label pieces of information. That's what variables do. Variables are great for all sorts of things – from tracking your score in a game to performing calculations and holding lists of items.



△ Storage box

A variable is like a box with a name label. You can store data in the box and then use the name to find the data again when you need to use it.

How to create a variable

A variable needs a name. Think of a name that will remind you what's inside the variable. Then decide what you want to store in the variable. This is the variable's value. Type the name, followed by an equals sign, followed by the value. We call this "assigning a value" to the variable.

1 Assign a value

In the shell window, type this line of code to create the variable **age** and assign a value to it. Use your own age if you want.

```
>>> age = 12
```

This value will be stored in the variable.

This is the variable's name.

2 Print the value

Now type the line of code shown on the right into the shell window. Hit the enter/return key to see what happens.

```
>>> print(age)
12
```

The value of **age**

The **print()** function prints the value of the variable between the brackets.



EXPERT TIPS

Naming variables

Choosing good names for your variables will make your program easier to understand. For example, a variable tracking a player's lives in a game could be called **lives_remaining**, rather than just **lives** or **lr**. Variable names can contain letters, numbers, and underscores, but they should begin with a letter. Follow the rules shown here and you won't go wrong.

Dos and don'ts

- Start the variable's name with a letter.
- Any letter or number can be used in the name.
- Symbols such as -, /, #, or @ aren't allowed.
- Spaces can't be used.
- An underscore (`_`) can be used instead of a space.
- Uppercase (capitals) and lowercase letters are different. Python will treat "Score" and "score" as two different variables.
- Avoid words Python uses as commands, such as "print".